Dear Editors:

We would like to submit the enclosed manuscript entitled "Variety Absorbs Variety: Workforce Skill Diversity, Shock Absorption and Youth Employment in Chinese Listed Firms" for consideration in Systems. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

When demand falls, young workers are the first to be let go. Yet some firms shed almost none of them, and the difference is not explained by size, leverage, profitability or sector. This manuscript asks what internal property of a firm decides whether an external disturbance is passed on to its youngest employees, and answers it with a principle that cybernetics established seventy years ago and that has never been tested on employment: Ashby's law of requisite variety, which holds that a system can absorb only as much variety as it contains. The urgency is not hypothetical. China produced a record 12.22 million graduates in 2025 while youth unemployment stayed above 17 percent, and the penalty for a poor start is measured in decades. Using 36,064 firm-year observations for 3,400 Chinese A-share listed firms over 2014–2024, four separate episodes of demand contraction and an industry shock constructed outside the firm, we show that the principle holds, and that it holds as a threshold.

The highlights of the paper are as follows.

(1) A measurable capacity, available before the shock arrives. Workforce skill variety is the entropy of a firm's employment distribution over five disclosed occupational categories and three education levels, so it is computable annually from public filings for every listed firm. It raises the youth employment share by 1.86 percentage points per standard deviation and attenuates the effect of an adverse demand shock, the interaction being 1.257. At one standard deviation below the mean of variety a shock costs 2.22 percentage points of youth employment; one standard deviation above it the effect is statistically indistinguishable from zero.

(2) The central result: absorption is a regime, not a gradient. Ashby's law is a sufficiency condition, and sufficiency conditions imply discontinuities. A fixed-effects panel threshold model locates one at 2.52 bits of variety, with a bootstrap p-value of 0.000 against the null of no threshold. Below it the shock cuts the youth employment share by 2.12 percentage points; above it the coefficient is −0.061 and insignificant. 48 percent of listed firm-years fall below the threshold, which is to say that for most firms the current workforce structure is not a weak buffer but no buffer at all.

(3) Which variety, and how it works. Related variety — depth within occupational families — buffers 1.49 times as strongly per bit as unrelated variety, the spread across families, because absorption requires that a person actually move and movement is governed by cognitive proximity. Of the buffering, 33 percent travels through suppressed separations and 14 percent through visible internal redeployment, so resilience is mostly a decision not to act. Breadth without depth looks like variety on an organization chart and is not variety in the regulatory sense.

(4) Identification and falsification. The disturbance is constructed from national industry revenue outside the firm. Variety is instrumented by the occupational diversity of the local labor pool and by the variety of same-industry firms in other provinces (Kleibergen–Paap F = 1345, Hansen p = 0.477), and the result survives propensity score matching, entropy balancing, seven robustness specifications, Oster bounds and a wild cluster bootstrap. The placebo is decisive: run on employees aged 31 to 50, whose separation is expensive, the buffering coefficient is 32 times smaller and insignificant. Variety protects exactly the margin the theory names.

We believe the manuscript is well suited to Systems. Its contribution is not the application of a systems vocabulary to an economic question but the empirical test of a systems law: we take a proposition from cybernetics, derive from it a prediction that no adjacent literature makes — that absorption requires a sufficient and therefore estimable quantity of internal variety — and find the threshold where the theory says it should be. In doing so the paper gives the organizational resilience literature something it has lacked, an antecedent of resilience that is observable before the shock rather than inferred from the recovery, and it gives the labor economics of adjustment an explanation for the dispersion it has documented but not accounted for. The practical payoff is unusually direct: the capacity is computed from disclosures every listed firm already files, the threshold is a number a human resource function can be held to, and the finding that the return to raising variety is concentrated at the crossing rather than spread along a gradient changes what an employment policy aimed at firms should try to buy. We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to the address below.

Thanks very much for your attention to our paper.

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Very sincerely yours,
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